

Education

10/2021 - 02/2026

Master of Science in Applied Geosciences (Grade: 2,5) - Karlsruhe Institute of Technology, Germany | [Certificate](#)
Thesis: Impact of Climate Change on the Geothermal Potential of Closed Systems Using GIS and Python | [Reference](#)

- Developed semi-analytical models in Python to assess shallow geothermal potential over 100 years to reduce carbon emissions and improve energy transition using heat pumps in different climate change scenarios.
- Converted modelling workflows from MATLAB to Python, improving real-time data integration for analysis.
- Integrated CMIP6 climate model data scenarios (SSP2-4.5 & SSP 5-8.5) from Google Earth Engine to integrate ground surface temperature and future geothermal conditions in a Python model.
- Created QGIS-based spatial visualizations & interactive dashboard maps using Folium based on Leaflet.
- Investigated how climate change affects geothermal systems with relevance for sustainable energy planning.

Volunteering: International Student Support for AGW & Buddy Program | [Certificate](#)

Scholarships: Stipendienprogramm-Gesetzes (StipG) by KIT Stiftung and e-fellows.net | [Certificate](#)

10/2020 - 10/2021

Master of Technology in Applied Geology (Drop out) - National Institute of Technology Raipur, India | [Certificate](#)
Project: Geological Mapping of Nithora Village, Chhattisgarh Using Remote Sensing and GIS Techniques | [Reference](#)

- Conducted geological mapping, including outcrop identification, rock-unit mapping, and structural data interpretation.
- Prepared geological maps and cross-sections using QGIS and CorelDraw, while applying image processing, topographic analysis, lithological classification, and survey methods to study the local geology.

Volunteering: National Service Scheme and Campus Ambassador of Protolith'20 | [Certificate](#)

Scholarship: ONGC Meritorious Scholarship by ONGC | [Certificate](#)

09/2017 - 09/2020

Bachelor of Science (Hons) in Geology (Grade 1,9) - Babasaheb Bhimrao Ambedkar University, India | [Certificate](#)
Thesis: Role of Meteorological Factors on Aerosol Load (PM_{2.5}) at Lucknow, India | [Reference](#)

- Conducted an environmental geoscience project on PM_{2.5} aerosol load, analyzing meteorological factors and spatio-temporal variations of particulate matter using air-quality and climate-related datasets.
- Applied data visualization and mapping tools, including WRPLOT, ArcGIS, QGIS, and CorelDraw, to interpret relationships between meteorological conditions and aerosol concentration patterns.

Volunteering: Student Campus Partner at Opportunity Cell, GeoYouth Fest & Foundation Day at BBAU | [Certificate](#)

Geological Field Excursions: Jabalpur, Ajmer, Solan | [Certificate](#)

Work Experience

07/2023 - 02/2026

Working Student - IONOS SE, Karlsruhe | [Reference](#)

- Developed web templates using HTML, CSS, and JavaScript for campaigns in a multi-tech environment.
- Applied AI integration (Claude & Copilot) with MCP servers to accelerate code generation, improve code quality, and support live testing, debugging, and faster implementation of frontend template changes.

01/2023 - 04/2023

Working Student - Hydrosion GmbH, (Bruchsal Geothermal Power Plant - EnBW) Munich, Germany | [Reference](#)

- Conducted hydrogeochemical investigations & sampling from deep geothermal, data processing, and analysis for direct lithium exploration from Geothermal brine on [Li+Fluids](#) and [UnLimited](#) projects.
- Performed geospatial data analysis in QGIS to support geothermal systems assessment across Germany.
- Supported technical documentation, reporting consistency, and internal communication across project teams.
- Assisted with documentation and environmental compliance support related to the JORC study.
- Contributed to process coordination, equipment logistics, and stock management with project partners.

05/2022 - 12/2022

Student Assistant - Institute of Applied Geosciences, Karlsruhe Institute of Technology, Germany | [Reference](#)

- Performed laboratory analyses on borehole heat exchanger and grouting material samples, including thermal conductivity, porosity, saturation, grain density, and wet bulk density testing.
- Designed and digitized 40+ maps, diagrams, and scientific visual assets using CorelDraw.
- Structured, documented, and evaluated 100+ datasets using Excel and Word.
- Petrographic and thin section analysis using microscopy and ImageJ for grain size & textural characterization.

Internship and Research Projects

07/2021 - 08/2021

Summer Training - Birbal Sahni Institute of Palaeosciences, Lucknow, India | [Certificate](#)

Title: Petrographical Characterization of the Lignite Deposits of Mangrol, Cambay Basin, Gujarat, India | [Reference](#)

- Conducted petrographical characterization of lignite deposits, applying coal petrography and organic geochemistry concepts to understand the composition and properties of the deposits.
- Interpreted depositional conditions and hydrocarbon potential of the Cambay Basin lignite deposits using sedimentary basin analysis and organic matter evaluation.

06/2021 - 07/2021

Summer Internship - Oil India Limited, Assam, India | [Certificate](#)

Title: Overview of Geology and Reservoir Engineering Concepts

- Gained practical exposure to geophysical survey methods, reservoir engineering concepts, petrophysical properties, and coring techniques used in hydrocarbon exploration.
- Developed an understanding of how geological, petrophysical, and engineering data are combined to evaluate reservoir quality and support hydrocarbon reservoir assessment.

06/2021 - 07/2021

Industrial Training - Oil and Natural Gas Corporation, India | [Certificate](#)

Title: A Preview of the Tools & Techniques Used for Hydrocarbon Exploration in Chattisgarh Basin, India

- Studied surface and subsurface exploration techniques used in frontier basins, including geological mapping, subsurface interpretation, and basin evaluation workflows.
- Gained exposure to well logging, drilling operations, rig systems, and hydrocarbon exploration methods used to assess subsurface resources in the Chhattisgarh Basin.

06/2019 - 06/2019

Summer Internship - Kumaun University, Nainital, India | [Certificate](#)

Title: Petrography to petrogenesis of granitoids | [Reference](#)

- Prepared rock thin sections and analyzed mineral optical properties using a light microscope to identify mineral assemblages and textural relationships in granitoid samples.
- Applied IUGS classification principles and petrographic interpretation methods to support granitoid classification and understand petrogenetic processes.

Skills

Technical Skills: Geospatial Data Analysis & Visualization, GIS & Geological Mapping, Climate Change, Shallow Geothermal & Energy Transition, CO₂ Emission & Decarbonization Assessment, Geodata Processing, Subsurface Drilling & Logging, Hydrogeochemical Sampling & Analysis, 3D Numerical Modelling, Hydrochemical Investigation

Programming & Tools: Python, SQL, Matlab, JavaScript, HTML, CSS, AI & Agents (ChatGPT, Claude code, Copilot)

Software: QGIS, Google Earth Engine, Leapfrog, CorelDraw, MS Office 365, Power BI, GitHub, WR Plot, EasyCore

Soft Skills: Strong Communication, Attention to Detail, Fast Learner, Analytical Mindset, Problem Solving, Structured

Language Skills: English (C1, Fluent), German (A2, Currently improving), Hindi (Mother tongue)

Certifications

- Fundamentals of Sustainability and Technology - IBM | [Certificate](#)
- Leapfrog: Geological Modelling & Importing Drilling and Surface Data - Seequent | [Certificate](#)
- Python for Data Science & Machine Learning with Python - IBM | [Certificate](#)
- Data Fundamentals & SQL and Relational Databases - IBM | [Certificate](#)
- Matlab - MINT-Kollegs Baden-Württemberg, KIT | [Certificate](#)
- Introduction to Remote Sensing - NPTEL | [Certificate](#)

Geological Field Excursions

- Central Alps fieldwork - Austria, 3 days | 08/2024
- Structural mapping field - Sammann & Knittlingen, 3 days | 2024
- Regional geology field - Derdinger Horn, Ottenau, Oberstrot, and Überlingen, 3 days | 2024
- Advanced geological mapping - Naxos, Greece, 10 days | 2022
- Reservoir Geological field school - Knittlingen, Oberderdingen, Gaggenau, and Baden-Baden, 5 days | 2022
- Geothermal plant and drilling site - Pullach, Holzkirchen, and Graben-Neudorf, Rittershoffen (Alsace) | 2022
- Eduard Merkle GmbH & Co. KG's limestone quarry and Hydroisotop GmbH - 2 days | 2022